Practice Programs

1. Matrix Multiplication

Your Code

```java
import java.util.Scanner;
public class main{
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        if(!sc.hasNextInt()) return;
        int r1=sc.nextInt();
        int c1=sc.nextInt();
        int[][] a=new int[r1][c1];
        for (int i=0;i<r1;i++){
            for(int j=0;j<c1;j++){
                a[i][j]=sc.nextInt();
            }
        }

        int r2=sc.nextInt();
        int c2=sc.nextInt();
        int[][] b= new int[r2][c2];
        for(int i=0;i<r2;i++){
            for(int j=0;j<c2;j++){
                b[i][j]=sc.nextInt();
            }
        }
        int[][] c=new int[r1][c2];
        for(int i=0;i<r1;i++){
            for(int j=0;j<c2;j++){
                int sum=0;
                for(int k=0;k<c1;k++){
                    sum+=a[i][k]*b[k][j];
                }
                c[i][j]=sum;
            }
        }

        for(int i=0;i<r1;i++){
            for(int j=0;j<c2;j++){
                System.out.print(c[i][j]);
                if(j<c2-1) System.out.print(" ");

            }
            if (i<r1-1) System.out.println();
        }
    }
}
```

Input

```
2 2
5 6
7 8
2 2
5 6
3 1
```

Output

```
43 36
59 50
```

2. Largest Number

Your Code

```java
import java.util.Scanner;
public class largestnum{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter size of array:");
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int largest=a[0];
        for(int i=0;i<n;i++){
            if(a[i]>largest){
                largest=a[i];
            }
        }
        System.out.println("largest element:"+largest);
    }
}

```

Input

```
4
25 41 30 12
```

Output

```
Enter size of array:
largest element:41
```

3. Smallest Number

Your Code

```java
import java.util.Scanner;
public class smallest{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
            int n=sc.nextInt();
            int[] a=new int[n];
            for(int i=0;i<n;i++){
                a[i]=sc.nextInt();
            }
            int small=a[0];
            for(int i=0;i<n;i++){
                if(a[i]<small){
                    small=a[i];

                }
            }
            System.out.print("Smallest:"+small);

    }
}
```

Input

```
4
5 9 12 56
```

Output

```
Smallest:5
```

4. Nth Largest Number

Your Code

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int N=sc.nextInt();
        for(int i=0;i<n-1;i++){
            for(int j=i+1;j<n;j++){
                if(a[i]<a[j]){
                    int temp=a[i];
                    a[i]=a[j];
                    a[j]=temp;
                }
            }
        }
        if(N<n)
            System.out.print("Nth largest number is: "+a[N+1]);
        else
            System.out.println("Invalid");
    }
}
```

Input

```
5
2 7 9 6 5
2
```

Output

```
Nth largest number is: 5
```

5. Palidrome Number/string

Your Code

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int  temp=n,rev=0;
        while(n!=0){
            int d=n%10;
            rev=rev*10+d;
            n=n/10;
        }
        if(temp==rev){
            System.out.println("Number is palindrome");
        }
        else{
            System.out.println("Number is not palindrome");
        }
        sc.nextLine();
        System.out.print("Enter a string:\n");
        String str=sc.nextLine();
        String r="";
```

Input

```
121
mam
```

Output

```
Number is palindrome
Enter a string:
String is palindrome
```

```
        }
        sc.nextLine();
        System.out.print("Enter a string:\n");
        String str=sc.nextLine();
        String r="";
        for(int i=str.length()-1;i>=0;i--){
            r+=str.charAt(i);
        }
        if(str.equals(r)){
            System.out.println("String is palindrome");
        }
        else{
            System.out.println("Not palindrome");
        }
        sc.close();
    }
}
```

6. Remove duplicates in Array

Your Code

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
            Scanner sc=new Scanner(System.in);
            int n=sc.nextInt();
            int[] a=new int[n];
            for(int i=0;i<n;i++){
                a[i]=sc.nextInt();
            }
            for(int i=0;i<n;i++){
                boolean duplicate=false;
                for(int j=0;j<i;j++){
                    if (a[i]==a[j]){
                        duplicate=true;
                        break;
                    }
                }
                if(!duplicate){
                    System.out.print(a[i]+" ");
                }
            }

    }
}
```

Input

```
4
2 5 2 4
```

Output

```
2 5 4
```